

Biochemistry 2013 Introduction to Medicine

Comprehensive Question Bank & Study Guide

*44 Detailed Molecular Biology, Genetics, & Metabolism Questions
with Comprehensive Explanations*

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1 Principles of Genetics & Classical Inheritance

Question 1

_____ is all DNA sequences required for synthesis of particular RNA transcript.

- A. Chromosome
- B. Gene
- C. Alleles
- D. Locus
- E. Chromatid

Correct Answer: B. Gene

In structural terms, a **gene** is defined as the fundamental unit of heredity that encompasses the DNA sequence encoding the structural components of a gene product (whether it be a polypeptide chain or an RNA molecule) along with the adjacent regulatory regions that control its expression.

Question 2

Which one of the following is incorrect about the genetic diseases?

- A. Genetic disease is any of the diseases and disorders that are caused by mutations in one or more genes.
- B. There are about 6000 known genetic disorders; all are fatal or cause severe problems.
- C. Monogenic disorders is inherited disorder due to defect on single gene that produce large effect
- D. Mitochondrial genetic inheritance disorder is one example of non-classic mendelian inheritance patterns in monogenic disorders
- E. All are correct

Correct Answer: B. There are about 6000 known genetic disorders; all are fatal or cause severe problems.

Many genetic mutations are mild, benign, or clinically asymptomatic under normal environmental conditions. For example, heterozygous carriers of autosomal recessive disorders (e.g., individuals with sickle cell trait or β -thalassemia minor) typically do not exhibit severe physical symptoms and have normal life expectancies. Evolutionary pressure can even maintain such mutations as polymorphisms when they confer a survival advantage (e.g., increased resistance to *Plasmodium falciparum* malaria).

Question 3

A large, three generation family in whom multiple members are affected with a rare, undiagnosed disease is being studied. Affected males never produce affected children, but affected females do produce affected children of both sexes when they mate with unaffected males. What is the most likely mode of inheritance?

- A. Autosomal dominant, with expression limited to females
- B. Y-linked
- C. Mitochondrial genetic inheritance disorders
- D. X-linked dominant
- E. X-linked recessive

Correct Answer: C. Mitochondrial genetic inheritance disorders

Mitochondrial DNA (mtDNA) is inherited strictly from the mother because mitochondria present in the sperm cell do not enter the fertilized egg during fertilization. Consequently, an affected mother will pass the mutation and the associated disorder to all of her offspring of both sexes. Affected males never transmit mitochondrial diseases to their children.

Question 4

A man who is affected with hemophilia A (X-linked recessive) mates with a woman who is a heterozygous carrier of this disorder. What proportion of this couple's daughters will be affected, and what proportion of the daughters will be heterozygous carriers, respectively?

- A. 0%; 50%
- B. 100%; 0%
- C. 0%; 100%
- D. 50%; 50%
- E. 2/3; 1/3

Correct Answer: D. 50%; 50%

Let X^h represent the mutant hemophilia A allele and X^H represent the normal allele.

- Affected father: X^hY (he can only pass his mutant X^h chromosome to his daughters).
- Carrier mother: $X^H X^h$ (she has a 50% chance of passing her normal X^H and a 50% chance of passing mutant X^h).

By crossing these gametes, the possible genotypes of their daughters are:

- $X^H X^h$ (Heterozygous Carrier): 50% probability.
- $X^h X^h$ (Homozygous Affected): 50% probability.

Question 5

A man and women are both affected by an autosomal dominant disorder. They are both heterozygotes for the disease-causing mutation. What is the probability that they will produce phenotypically normal offspring?

- A. 50%
- B. 25%
- C. 75%
- D. 100%
- E. None

Correct Answer: B. 25%

Let D represent the dominant disease-causing allele and d represent the normal recessive allele. Both parents are heterozygous ($Dd \times Dd$). The Punnett square of this cross yields the following offspring genotypic ratios:

- 25% DD (homozygous affected)
- 50% Dd (heterozygous affected)
- 25% dd (homozygous recessive normal)

Since the disorder is dominant, only the dd genotype will be phenotypically normal, which corresponds to a **25%** probability.

Question 9

Which of one of the following is true regarding to X-linked disorders?

- A. If heterozygous female met normal male; 100% of female off-springs suffers from disorders

- in X-linked dominant disorders
- B. If hemizygous male met normal female; 100% of male off-springs suffers from disorders
 - C. If hemizygous male met heterozygous female; 50% of male off-springs have normal genotype and phenotype
 - D. If heterozygous female met normal male; 100% of female off-springs suffer from disorder in X-linked recessive disorders
 - E. None

Correct Answer: C. If hemizygous male met heterozygous female; 50% of male off-springs have normal genotype and phenotype

Let the affected male be X^hY and the carrier female be $X^H X^h$. Male offspring inherit their Y chromosome from the father and either the X^H or X^h chromosome from the mother. There is a 50% chance of inheriting the normal X^H chromosome, giving them a normal genotype and phenotype ($X^H Y$). Hence, 50% of male offspring will be completely normal.

Question 11

Somatic cells reproduce by _____, while sex cells reproduce by _____.

- A. Meiosis; mitosis
- B. Mitosis; mitosis
- C. Mitosis; meiosis
- D. Meiosis; meiosis

Correct Answer: C. Mitosis; meiosis

Somatic (body) cells utilize **mitosis** to replicate, creating genetically identical diploid cells for tissue growth and repair. In contrast, germ cells in the gonads undergo **meiosis** to produce haploid gametes (sperm and egg cells) containing half the parental genetic complement.

Question 12

When _____ occurs between nonsister chromatids, genetic exchange between chromosomes provides new combination of genes that are different from either parent.

- A. Cytokinesis
- B. Crossing-over
- C. Mitosis cell division
- D. All

Correct Answer: B. Crossing-over

Crossing-over is a process of homologous recombination that takes place during prophase I of meiosis. Nonsister chromatids of homologous chromosomes pair up (synapsis) and physically exchange segments of DNA. This process breaks old genetic linkages and produces recombinant chromatids, resulting in offspring with unique genetic combinations distinct from either parent.

Question 13

In mitosis, _____ phase begins when pairs of sister chromatids align in the center of the cell; and _____ phase is associated with the formation of the nuclear envelope, respectively.

- A. Anaphase; Metaphase
- B. Metaphase; Telophase
- C. Interphase; Prophase
- D. Prophase; Anaphase

Correct Answer: B. Metaphase; Telophase

- **Metaphase** begins as chromosomes condense fully and migrate to the equator of the cell (metaphase plate), where their kinetochores attach to the spindle microtubules.
- **Telophase** is the final stage of mitosis where segregated chromosomes reach opposite poles, the mitotic spindle deassembles, and a new nuclear envelope reforms around each set of daughter chromosomes.

2 Cell Cycle Control & Molecular Regulation

Question 14

At the _____ checkpoint, cell growth is controlled.

- A. G1
- B. G2
- C. S
- D. None of the above

Correct Answer: A. G1

The **G1 checkpoint** (restriction point) is the primary site of cell growth control. Here, the cell monitors its size, nutrient levels, growth factor signaling, and DNA integrity before deciding whether to commit to the cell cycle and enter S-phase (DNA replication).

Question 15

Microtubules grow from the poles to kinetochores of each chromatid during _____.

- A. Prophase
- B. Metaphase
- C. Anaphase
- D. Telophase

Correct Answer: A. Prophase (specifically prometaphase/late prophase)

During **prophase**, chromatin condenses into distinct chromosomes and the centrosomes migrate to opposite poles to establish the spindle apparatus. Microtubules nucleate from the centrosomes and grow toward the kinetochores on the condensed sister chromatids to stabilize and position them.

Question 16

P 53:

- A. Proto-oncogene
- B. An oncogene
- C. Control progression of cell cycle from G1-phase to S-phase
- D. The action of p53 is induced in response to impairment of replicated DNA molecules.

Correct Answer: D. The action of p53 is induced in response to impairment of replicated DNA molecules.

The protein **p53** acts as the "guardian of the genome". When DNA damage occurs (e.g., from UV radiation, reactive oxygen species, or mutagens), p53 degradation is halted, leading to its accumulation. Active p53 functions as a transcription factor, driving the expression of p21 (which arrests the cell cycle in G1/S phase to allow for repair), GADD45 (which promotes DNA repair), or apoptotic proteins (like Bax) if repair fails. It is a tumor-suppressor gene, not an oncogene or proto-oncogene.

Question 17

Which one of the following statement is correct about cyclin and Cdks?

- A. The concentration of cyclin doesn't fluctuate during the cell cycle
- B. S-cyclin stimulates chromosome duplication & contributes to control of some early mitotic events.
- C. To modify target proteins Cdk shouldn't bind with cyclin.
- D. Cdks are not constant as cell progresses through cycle

Correct Answer: B. S-cyclin stimulates chromosome duplication & contributes to control of some early mitotic events.

Cyclin-dependent kinases (CDKs) are present at constant levels throughout the cell cycle but remain catalytically inactive unless bound by their regulatory partners, the cyclins. Cyclin levels fluctuate dynamically. S-phase cyclins bind CDKs during G1/S to form complexes that phosphorylate key substrates, initiating DNA replication (chromosome duplication) and setting up subsequent mitotic events.

Question 42

In cell cycle, the rate of transcription & translation except for Histone proteins is at its lowest level during _____

- A. G1-Phase of interphase
- B. S-phase of interphase
- C. G2-Phase of interphase
- D. Mitotic Phase

Correct Answer: D. Mitotic Phase

During the **Mitotic (M) phase**, chromatin undergoes massive condensation to ensure the precise partition of chromosomes into two daughter cells. This highly compacted structure restricts access of RNA polymerases and transcription factors to the DNA template, shutting down global gene transcription and translation.

3 Free Radicals, Oxidative Stress & Antioxidants

Question 18

Which one of the following causes free radicals to form that potentially putting you at greater risk of heart disease?

- A. High fat foods
- B. Taking high amounts of a single antioxidant
- C. Exercise
- D. All of the above
- E. None of the above

Correct Answer: A. High fat foods

Diets rich in fats increase the delivery of electron donors (NADH and $FADH_2$) to the mitochondrial electron transport chain. When the capacity of the respiratory chain is saturated, electrons leak directly to oxygen, forming the superoxide free radical ($O_2^{\bullet-}$). Accumulation of free radicals leads to the oxidative modification of circulating low-density lipoproteins (LDL), which promote atherosclerosis and put individuals at a higher risk of heart disease.

Question 19

Which of the following statements is true as it relates to your genes and free radical activity in your body?

- A. Free radicals promote gene and cell division.
- B. Free radicals can cause genetic mutations.
- C. Free radicals cause cancer.
- D. A and C are correct.
- E. B and C are correct.

Correct Answer: E. B and C are correct.

Free radicals are highly reactive species that can attack DNA, causing chemical modifications such as single- or double-strand breaks and the oxidation of bases (e.g., creating 8-oxoguanine, which mispairs with adenine). If these lesions escape DNA repair, they lead to stable **genetic mutations**. Accumulation of mutations in key growth-control genes (proto-oncogenes and tumor suppressor genes) is a driver for the initiation and progression of **cancer**.

Question 20

Under special circumstances, free radicals can be a good thing for your body because:

- A. They can give you a boost of energy at the end of a long distance workout.
- B. They are the weapons your body uses to fight off viruses and bacteria.
- C. They can protect your skin from the sun's harmful rays.
- D. They cause mutations in our genetic code, and without that our species wouldn't evolve

Correct Answer: B. They are the weapons your body uses to fight off viruses and bacteria.

During the immune response, phagocytic cells (neutrophils and macrophages) utilize the enzyme complex NADPH oxidase to reduce molecular oxygen into superoxide free radicals. This respiratory burst is localized within phagolysosomes, generating toxic reactive oxygen intermediates to destroy phagocytosed bacteria and viruses.

Question 21

Which of the following statement is correct regarding antioxidant?

- A. They present in nearly the same concentrations relative to the oxidants
- B. They help repair damage already sustained by cells.
- C. SOD, catalase and glutathione are example of enzymatic antioxidants.
- D. All of the above

Correct Answer: C. SOD, catalase and glutathione are example of enzymatic antioxidants.

Superoxide dismutase (SOD), catalase, and glutathione peroxidase are the primary enzymatic defense systems that convert reactive oxygen species into non-toxic substances (such as converting superoxide to hydrogen peroxide, and then to water). Non-enzymatic antioxidants are typically present at far lower concentrations than potential cellular targets, and their primary function is to prevent free radical propagation rather than repair molecular damage that has already occurred.

Question 22

Which of the following statements is true concerning lipid peroxidation?

- A. It refers to oxidative degradation of DNA, lipids and proteins.
- B. The process proceeds by free radical chain reaction mechanism.
- C. It most often affects saturated fatty acids.
- D. End products of lipid peroxidation are not mutagenic and carcinogenic.

Correct Answer: B. The process proceeds by free radical chain reaction mechanism.

Lipid peroxidation is a destructive process that specifically degrades lipids. The reaction proceeds through a classic free radical chain reaction mechanism (comprising initiation, propagation, and termination steps). It primarily affects **unsaturated** fatty acids (which contain reactive double bonds), leading to membrane damage and generating toxic, mutagenic, and carcinogenic end-products such as malondialdehyde.

Question 23

Which one of the following cellular processes does not generate free radicals?

- A. Mitochondrial electron transport
- B. Inflammation
- C. During the normal oxidation of foodstuffs
- D. None of the above

Correct Answer: D. None of the above

All of the listed options generate free radicals:

- **Mitochondrial electron transport** regularly leaks electrons to oxygen to form superoxide.
- **Inflammation** stimulates phagocytes to generate massive amounts of free radicals via NADPH oxidase to kill pathogens.
- **Normal oxidation of foodstuffs** continuously transfers electrons through metabolic pathways, yielding reactive oxygen intermediates as metabolic by-products.

Question 24

Which of the following statement is not appropriately describing free radical nitric oxide?

- A. Helps blood vessels to dilate
- B. Acts as a chemical messenger in the brain.
- C. Brings about bactericidal action.
- D. None of the above

Correct Answer: D. None of the above

All of the statements are excellent and appropriate descriptions of the nitric oxide ($NO^\bullet$) free radical:

- It is synthesized by endothelial cells and diffuses into vascular smooth muscle to activate guanylyl cyclase, causing relaxation and **vasodilation**.
- It acts as a specialized **chemical messenger (neurotransmitter)** in the central nervous system.
- In macrophages, it combines with superoxide to form peroxynitrite, a powerful oxidant that mediates **bactericidal action**.

4 DNA Replication, Repair & Recombinant DNA

Question 25

Which enzyme is used to join nicks in the DNA strand?

- A. Primase
- B. DNA polymerase
- C. DNA ligase
- D. Endonuclease

Correct Answer: C. DNA ligase

DNA ligase catalyzes the formation of a phosphodiester bond between the 3'-hydroxyl group of one DNA fragment and the 5'-phosphate group of another, effectively sealing nicks or gaps in the phosphodiester backbone of double-stranded DNA during replication (joining Okazaki fragments) and repair pathways.

Question 26

Short strands of _____-primer is used in DNA replication.

- A. DNA
- B. RNA
- C. Histone
- D. Protein

Correct Answer: B. RNA

DNA polymerases are unable to initiate DNA synthesis *de novo* on a single-stranded template; they can only add nucleotides to a pre-existing 3'-hydroxyl terminus. This necessary free 3'-OH group is supplied by a short complementary **RNA primer** synthesized by the enzyme primase.

Question 31

Restriction endonucleases

- A. Cut RNA chains at specific locations
- B. Excise introns from hnRNA
- C. Remove Okazaki fragments
- D. Act as defensive enzymes to protect the host bacterial DNA from DNA of foreign organisms

Correct Answer: D. Act as defensive enzymes to protect the host bacterial DNA from DNA of foreign organisms

In nature, restriction endonucleases are bacterial enzymes that cleave foreign double-stranded DNA molecules at specific palindromic recognition sequences. This bacterial restriction-modification system serves as a defensive mechanism to destroy foreign bacteriophage DNA while host DNA is protected by methylation.

Question 32

All the following statements about xeroderma pigmentosum are true except

- A. It is a genetic disease
- B. Its inheritance is autosomal dominant
- C. uvr ABC excinuclease is defective in this disease
- D. It results in multiple skin cancers

Correct Answer: B. Its inheritance is autosomal dominant

Xeroderma pigmentosum (XP) is inherited as an **autosomal recessive** genetic disease, not autosomal dominant. It is caused by mutations in genes coding for proteins involved in nucleotide excision repair (NER). A failure in this pathway prevents cells from repairing UV-induced pyrimidine (thymine) dimers, leading to severe sun sensitivity, skin abnormalities, and multiple skin cancers.

Question 33

All of the following may be used as expression vectors except

- A. Plasmid
- B. Bacteriophage
- C. Cosmids
- D. E. coli

Correct Answer: D. E. coli

In recombinant DNA technology, plasmids, bacteriophages, and cosmids are DNA molecules used as **vectors** to transport foreign genetic material into a host cell. *Escherichia coli* (*E. coli*) is a host bacterium inside of which these vectors are introduced and replicated; it is not a vector itself.

Question 40

The failure of _____ repair predisposes to HNPCC (hereditary non-polyposis colorectal cancer) development.

- A. Mismatch
- B. Double strand break
- C. Base excision
- D. B & C

Correct Answer: A. Mismatch

Hereditary nonpolyposis colorectal cancer (HNPCC, or Lynch syndrome) is a genetic condition caused by inherited mutations in genes (such as *MSH2* or *MLH1*) encoding proteins that are essential for the **mismatch repair** pathway. When mismatch repair is compromised, cells accumulate mutations rapidly in microsatellites and growth-regulatory genes, leading to oncogenesis.

5 Transcription & Post-Transcriptional Processing

Question 27

Which one of the following statement is incorrect about codon and anti-codon?

- A. Both codon and anti-codon consist of sequence of three nucleotides.
- B. Codons are present on DNA and anti-codon is present on tRNA.
- C. Codon and anticodon are complementary in nature.
- D. None of the above

Correct Answer: B. Codons are present on DNA and anti-codon is present on tRNA.

This statement is incorrect because **codons are present on messenger RNA (mRNA)**, not DNA. The mRNA codons are transcribed from the template DNA strand and are subsequently read by the tRNA anticodons during translation in the cytosol.

Question 28

The correct statement concerning RNA and DNA polymerases is

- A. RNA polymerase use nucleoside diphosphates.
- B. RNA polymerase requires primers and adds bases at 5' end of the growing polynucleotide chain.
- C. DNA polymerases can add nucleotides at both ends of the chain.
- D. All RNA and DNA polymerases can add nucleotides only at the 3' end of the growing polynucleotide chain

Correct Answer: D. All RNA and DNA polymerases can add nucleotides only at the 3' end of the growing polynucleotide chain

Both RNA and DNA polymerases synthesize polynucleotide chains exclusively in the **5' to 3' direction**. They catalyze the attack of the 3'-hydroxyl group of the growing chain on the 5'-triphosphate of the incoming nucleotide, meaning bases can only be added to the 3' end.

Question 29

5'-Terminus of mRNA molecule is capped with

- A. Guanosine triphosphate
- B. 7-Methylguanosine triphosphate

- C. Adenosine triphosphate
- D. Adenosine diphosphate

Correct Answer: B. 7-Methylguanosine triphosphate

Eukaryotic mRNA is modified co-transcriptionally at its 5' end by the addition of a **7-methylguanosine cap** via an unusual 5' to 5' triphosphate linkage. This cap protects the mRNA from exonuclease degradation and serves as a critical recognition signal for the translation initiation machinery.

Question 36

The enzyme that violates the central dogma of flow of genetic information & also contributes to resist chromosomal erosion is....

- A. Primase
- B. DNA polymerase
- C. Reverse Transcriptase
- D. RNA polymerase

Correct Answer: C. Reverse Transcriptase

The classic central dogma dictates information flow from DNA → RNA → protein. Reverse transcriptases violate this by synthesizing DNA from an RNA template. **Telomerase** is a specialized ribonucleoprotein complex containing reverse transcriptase activity that utilizes its own internal RNA molecule as a template to extend the 3' end of linear chromosomes, preventing chromosomal erosion.

Question 41

In prokaryotic transcription the promoter regions(-35 & -10 consensus sequences) are recognized by

- A. α -factor
- B. β -subunit
- C. σ -subunit
- D. TF-IIID

Correct Answer: C. σ -subunit

In prokaryotes, the core RNA polymerase enzyme ($2\alpha, 1\beta, 1\beta', 1\omega$) must associate with a specialized regulatory protein called the **sigma (σ) factor** (subunit) to form the holoenzyme. The σ subunit directly recognizes and binds to the -10 (Pribnow box) and -35 promoter consensus sequences on the DNA template, guiding the polymerase to initiate transcription at the proper start site.

Question 43

The antibiotic Rifampin is used in the treatment of Tuberculosis (TB), this antibiotic works by

- A. Inhibiting α -factor of RNA polymerase
- B. Inhibiting the β -subunit of RNA polymerase
- C. Inhibiting RNA polymerase-II
- D. Inhibiting the β' -subunit of RNA polymerase

Correct Answer: B. Inhibiting the β -subunit of RNA polymerase

Rifampin is a first-line bactericidal drug that binds specifically to the **β subunit of prokaryotic RNA polymerase**. This physical blockade prevents the enzyme from forming the first phosphodiester bond, thereby selectively arresting bacterial transcription initiation without affecting eukaryotic nuclear RNA polymerases.

6 Translation & Protein Synthesis

Question 10

The synthesis of all proteins in both prokaryotic and eukaryotic cells begins with which of the following amino acids?

- A. Alanine
- B. Glycine
- C. Methionine
- D. All

Correct Answer: C. Methionine

The universal start (initiation) codon is 5'-AUG-3'. During translation initiation, AUG recruits a specialized initiator tRNA charged with **methionine**. In eukaryotes, this amino acid is unmodified methionine, whereas in prokaryotes, it is N-formylmethionine.

Question 30

Initiation of protein synthesis begins with binding of

- A. 40S ribosomal unit on mRNA
- B. 60S ribosomal unit
- C. Charging of tRNA with specific amino acid
- D. Attachment of aminoacyl tRNA on mRNA

Correct Answer: A. 40S ribosomal unit on mRNA

In eukaryotic translation, protein synthesis initiates when the small **40S ribosomal subunit** (pre-associated with initiation factors and met-tRNA) binds to the 5'-methylguanosine cap of the mRNA. The 40S subunit then scans the 5'-untranslated leader sequence in a 5'-to-3' direction until it identifies the AUG start codon, after which the large 60S subunit joins to form the functional 80S ribosome.

Question 34

You are studying the processes of protein synthesis via the use of a cell-free system and synthetic RNAs. You have introduced the following RNA to this system: 5'-AUAUAAUGACUAAAAUAU-3' Using the indicated codons below, which of the following is the most likely sequence of the peptide synthesized in this system? Codon: AAA (Lysine), AUG (Methionine), AAU (Asparagine), ACU (Threonine), AUA (Isoleucine), UAU (Tyrosine), CAU (Leucine)

- A. Isoleucine, methionine, threonine, lysine, tyrosine

- B. Methionine, isoleucine, asparagine
- C. Methionine, threonine, leucine, threonine
- D. Methionine, threonine, lysine, tyrosine

Correct Answer: D. Methionine, threonine, lysine, tyrosine

Translation does not begin at the 5'-terminal base of an mRNA; it must initiate at the first start codon, which is **5'-AUG-3'**. Analyzing the synthetic RNA sequence:

5'- A U A U A A [**AUG**] A C U A A A U A U -3'

The first AUG begins at nucleotide 7. Reading the downstream sequence in non-overlapping triplets (codons) from this start site:

1. **AUG** → Methionine (Met)
2. **ACU** → Threonine (Thr)
3. **AAA** → Lysine (Lys)
4. **UAU** → Tyrosine (Tyr)

This yields the peptide: **Methionine, threonine, lysine, tyrosine.**

Question 44

In Huntington's disease (a neurodegenerative disorder) is caused by the Trinucleotide Repeat Expansion of the codon _____ coding the amino acid _____ which results in the formation of Huntington's proteins which aggregates and damage brain cells.

- A. CAG, Glutamine
- B. CUG, Leucine
- C. CAG, Glutamate
- D. UCU, Serine

Correct Answer: A. CAG, Glutamine

Huntington's disease is a classic trinucleotide repeat expansion disorder. It is caused by the expansion of tandem **CAG** repeats within the coding region of the huntingtin (*HTT*) gene on chromosome 4. Because the **CAG** codon specifies the amino acid **Glutamine**, this mutation leads to an abnormally long polyglutamine tract, causing the huntingtin protein to misfold, aggregate, and selectively damage GABAergic neurons in the striatum.

7 Advanced Genomics & Structural Biochemistry

Question 35

Which of the following statement is odd regarding the applications of contemporary genomics?

- A. Drug Designing
- B. Drug Dispensing
- C. Crime Investigation
- D. Genealogy

Correct Answer: B. Drug Dispensing

Drug dispensing is a traditional, operational task of preparing and giving out medications in a pharmacy. In contrast, drug designing (pharmacogenomics and molecular modeling), crime investigation (forensic DNA profiling), and genealogy (ancestry and maternity/paternity tracing) represent direct applications of contemporary genomic and molecular biology technologies.

Question 37

In which of the following nucleoside, the N-glycosidic linkage is non-existent?

- A. Thymidine
- B. Pseudouridine
- C. Adenosine
- D. Uridine

Correct Answer: B. Pseudouridine

In standard nucleosides (such as thymidine, adenosine, and uridine), the nitrogenous base is attached to the C1' of the ribose sugar via an N-glycosidic (N-C) bond. In **pseudouridine** (ψ)—a modified nucleoside commonly found in transfer RNA—the uracil base is rotated and linked to the ribose C1' via a stable **carbon-carbon (C-C) bond**, making the N-glycosidic linkage non-existent.

Question 38

The sugar that is incorporated in anticancer drugs like Cytarabin & Vidarabin so that DNA replication can be evaded successfully in proliferating cells is...

- A. Ribose
- B. Xanthose

- C. Arabinose
- D. Erythrose

Correct Answer: C. Arabinose

Cytarabine (cytosine arabinoside) and vidarabine (adenine arabinoside) are nucleoside analogs that incorporate the pentose sugar **arabinose** instead of ribose or deoxyribose. When rapidly dividing cancer cells incorporate these nucleotide analogs into their DNA during replication, the stereochemical block from the 2'-hydroxyl group of arabinose prevents further chain elongation and halts DNA synthesis.

Question 39

In the organization of eukaryotic DNA, the histone protein that doesn't take part in the formation of histone octamer is...

- A. H2A
- B. H2B
- C. H1
- D. H4

Correct Answer: C. H1

Eukaryotic chromatin is organized into nucleosome core beads. Each nucleosome core consists of approximately 147 base pairs of DNA wrapped around a histone octamer formed by two molecules each of core histones **H2A, H2B, H3, and H4**. Histone **H1** is a linker histone that binds to the linker DNA between nucleosome beads to facilitate high-order packing and stabilization of the chromatin fiber.